Numerical Problems in Probability

for Computer Science

A worked-problem paper

Ninety problems with full solutions

Binomial, Poisson, and normal models; the two classical approximations; negative binomial and geometric waiting times; tests for one mean and two means

22 October 2026

Abstract

This paper collects ninety original numerical problems from the part of probability that computer scientists use most often: counting independent trials, counting rare arrivals, reading a normal table, approximating an awkward binomial probability, waiting until a trial succeeds, and deciding whether a measured mean has really changed. Each of the nine topics has ten problems. A full solution follows every problem. The final section is a one-line answer key, and the appendices print every numerical table the solutions rely on, so the paper can be read without a separate formula card.

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How to read the solutions

The problems are set in computing situations: packet delivery, caches, build times, replicas, queues, compilers, and benchmarks. The random model is always stated in the problem. Independence and a constant success probability are assumptions of the model, not facts the data have proved.

Rounding.

- A probability answer is rounded half up to four decimal places.
- Powers of e are taken from Appendix A, already rounded half up to six decimal places. Products of those values are then rounded to four decimal places at the end.
- A standard-normal probability is read from Appendix B after the z-score has been rounded half up to two decimal places.
- A t statistic is reported to three decimal places and compared with Appendix C.
- The normal critical values used for tests are 1.645, 1.960, 2.326, and 2.576. They are also the last row of the t table.

A hypothesis test is written in the same five lines every time. State H_0 and H_1, name α and the distribution, compute the statistic, compare it with the critical value, and write the conclusion in the language of the system being measured. A p-value is given as well. For a z test it comes from Appendix B. For a t test the printed table gives a range, which is enough for the decision.

What the four decimal places do not mean. A model probability such as 0.1937 is not a measured frequency. It is the value of the formula under the stated assumptions. If the trials are dependent, or the success probability drifts, the formula does not describe the system, however carefully it is rounded.

1 The binomial distribution

A binomial count is the number of successes in a fixed number of independent yes-or-no trials. It fits a packet that is either delivered or not, a test that either passes or fails, and a replica that either votes correctly or does not, provided the trials do not influence one another and they share one success probability.

Let X be the number of successes in n trials, with success probability p and failure probability $q = 1 - p$. Then $X \sim \mathrm{Bin}(n, p)$ and

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n. \tag{1}$$

The mean, variance, and standard deviation are

$$\mathrm{E}[X] = np, \qquad \mathrm{Var}(X) = npq, \qquad \mathrm{SD}(X) = \sqrt{npq}. \tag{2}$$

The complement is often the short way to an upper or lower tail:

$$P(X \geq 1) = 1 - P(X = 0), \qquad P(X \leq 1) = P(X = 0) + P(X = 1).$$

A retry loop that stops at the first success is not binomial, because the number of trials is random. That waiting time is geometric, and it is treated in Section 7.

Problem 1.1. Packet delivery

A transport prototype sends 10 packets. Each packet is delivered correctly with probability 0.9, independently of the others. Find the probability that exactly 8 packets are delivered correctly.

Solution. Here $n = 10$, $p = 0.9$, $q = 0.1$, and $k = 8$.

$$P(X = 8) = \binom{10}{8}(0.9)^8(0.1)^2.$$

$\binom{10}{8} = 45$ and $(0.1)^2 = 0.01$. Also $(0.9)^2 = 0.81$, so $(0.9)^4 = 0.6561$ and

$$(0.9)^8 = 0.6561^2 = 0.43046721.$$

Therefore

$$P(X = 8) = 45 \times 0.43046721 \times 0.01 = 0.1937102445,$$

which rounds to 0.1937.

Answer. About 19.37% of such batches contain exactly eight correct deliveries.

Problem 1.2. Unit tests

Eight independent unit tests each pass with probability 0.75. Find the probability that at least seven tests pass.

Solution. Let $X \sim \mathrm{Bin}(8, 0.75)$. The event $X \geq 7$ is the union of $X = 7$ and $X = 8$.

$$(0.75)^2 = 0.5625, \quad (0.75)^4 = 0.31640625, \quad (0.75)^8 = 0.100112915.$$

$$(0.75)^7 = (0.75)^8/0.75 = 0.133483887.$$

$$P(X = 7) = \binom{8}{7}(0.75)^7(0.25) = 8 \times 0.133483887 \times 0.25 = 0.266967774.$$

$$P(X = 8) = (0.75)^8 = 0.100112915.$$

$$P(X \geq 7) = 0.266967774 + 0.100112915 = 0.367080689,$$

which rounds to 0.3671.

Answer. The chance that at least seven of the eight tests pass is 0.3671.

Problem 1.3. Cache hits

A lookup hits the cache with probability 0.3. Twenty lookups are independent. Find the mean, variance, and standard deviation of the number of hits.

Solution. Let $X \sim \mathrm{Bin}(20, 0.3)$. Then $q = 0.7$ and

$$\begin{aligned} \mathrm{E}[X] &= 20 \times 0.3 = 6, \\ \mathrm{Var}(X) &= 20 \times 0.3 \times 0.7 = 4.2, \\ \mathrm{SD}(X) &= \sqrt{4.2} = 2.0494. \end{aligned}$$

Answer. The expected number of hits is 6, the variance is 4.2, and the standard deviation is 2.0494 hits.

Problem 1.4. Bit flips

Five bits are transmitted. Each bit is flipped with probability 0.2, independently. Find the probability that at most one bit is flipped.

Solution. Let $X \sim \mathrm{Bin}(5, 0.2)$ be the number of flipped bits. Then $q = 0.8$ and

$$P(X = 0) = (0.8)^5 = 0.32768,$$

$$P(X = 1) = \binom{5}{1}(0.2)(0.8)^4 = 5 \times 0.2 \times 0.4096 = 0.4096.$$

$$P(X \leq 1) = 0.32768 + 0.4096 = 0.73728,$$

which rounds to 0.7373.

Answer. The probability of fewer than two flipped bits is 0.7373.

Problem 1.5. API calls

Six independent calls are made to an API. Each call succeeds with probability 0.8. Find the probability that every call succeeds, and the probability that at least one call fails.

Solution. Let $X \sim \mathrm{Bin}(6, 0.8)$ be the number of successes.

$$(0.8)^2 = 0.64, \qquad (0.8)^4 = 0.4096, \qquad (0.8)^6 = 0.4096 \times 0.64 = 0.262144.$$

So $P(X = 6) = 0.262144$, which rounds to 0.2621. At least one failure is the complement:

$$P(X \le 5) = 1 - 0.262144 = 0.737856,$$

which rounds to 0.7379.

Answer. Every call succeeds with probability 0.2621. At least one call fails with probability 0.7379.

Problem 1.6. Spam messages

A mailbox contains 12 messages. Each message is spam with probability 0.25, independently. Find the probability that exactly three messages are spam.

Solution. Let $X \sim \mathrm{Bin}(12, 0.25)$. Then $\binom{12}{3} = 220$, $(0.25)^3 = 0.015625$, and

$$\begin{aligned}(0.75)^2 &= 0.5625,\\ (0.75)^4 &= 0.31640625,\\ (0.75)^8 &= 0.100112915,\\ (0.75)^9 &= 0.075084686.\end{aligned}$$

$$P(X = 3) = 220 \times 0.015625 \times 0.075084686 = 3.4375 \times 0.075084686 = 0.258103609,$$

which rounds to 0.2581.

Answer. The probability of exactly three spam messages is 0.2581.

Problem 1.7. Module defects

Five modules are reviewed independently. Each module contains a critical defect with probability 0.4. Find the probability that exactly two modules contain a critical defect.

Solution. Let $X \sim \mathrm{Bin}(5, 0.4)$. Then $\binom{5}{2} = 10$, $(0.4)^2 = 0.16$, and $(0.6)^3 = 0.216$.

$$P(X = 2) = 10 \times 0.16 \times 0.216 = 1.6 \times 0.216 = 0.3456.$$

The value 0.3456 is already at four decimal places.

Answer. The probability of exactly two defective modules is 0.3456.

Problem 1.8. Advertisement clicks

Fifteen users each click an advertisement with probability 0.2, independently. Find the expected number of clicks and the probability that nobody clicks.

Solution. Let $X \sim \mathrm{Bin}(15, 0.2)$.

$$\mathrm{E}[X] = 15 \times 0.2 = 3.$$

No clicks means every user refuses:

$$P(X = 0) = (0.8)^{15}.$$

Since $(0.8)^4 = 0.4096$ and $(0.8)^8 = 0.16777216$,

$$\begin{aligned}(0.8)^{12} &= 0.16777216 \times 0.4096 = 0.068719477,\\ (0.8)^{15} &= 0.068719477 \times 0.512 = 0.035184372.\end{aligned}$$

This rounds to 0.0352.

Answer. The expected number of clicks is 3. The probability of no clicks is 0.0352.

Problem 1.9. Replica votes

A commit protocol asks nine replicas. Each replica votes correctly with probability 0.7, independently. The commit is accepted only when at least five replicas are correct. Find the probability that the commit is accepted.

Solution. Let $X \sim \mathrm{Bin}(9, 0.7)$. The required probability is $P(X \geq 5)$. The binomial coefficients are

$$\binom{9}{5} = 126, \quad \binom{9}{6} = 84, \quad \binom{9}{7} = 36, \quad \binom{9}{8} = 9, \quad \binom{9}{9} = 1.$$

The powers needed are $(0.7)^5 = 0.16807$, $(0.7)^6 = 0.117649$, $(0.7)^7 = 0.0823543$, $(0.7)^8 = 0.05764801$, $(0.7)^9 = 0.040353607$, and $(0.3)^2 = 0.09$, $(0.3)^3 = 0.027$, $(0.3)^4 = 0.0081$.

$$\begin{aligned} P(X = 5) &= 126 \times 0.16807 \times 0.0081 = 0.171532, \\ P(X = 6) &= 84 \times 0.117649 \times 0.027 = 0.266828, \\ P(X = 7) &= 36 \times 0.0823543 \times 0.09 = 0.266828, \\ P(X = 8) &= 9 \times 0.05764801 \times 0.3 = 0.155650, \\ P(X = 9) &= 0.040354. \end{aligned}$$

Adding these five probabilities,

$$P(X \geq 5) = 0.171532 + 0.266828 + 0.266828 + 0.155650 + 0.040354 = 0.901192,$$

which rounds to 0.9012.

Answer. The commit is accepted with probability 0.9012.

Problem 1.10. Source files

Seven source files are compiled independently. Each file compiles with probability 0.6. Find the probability that the number of successful compilations is from 2 to 4, inclusive.

Solution. Let $X \sim \mathrm{Bin}(7, 0.6)$. The required event is $X \in \{2, 3, 4\}$.

$$\binom{7}{2} = 21, \qquad \binom{7}{3} = \binom{7}{4} = 35.$$

$$\begin{aligned} P(X = 2) &= 21 \times (0.6)^2 \times (0.4)^5 = 21 \times 0.36 \times 0.01024 = 0.077414, \\ P(X = 3) &= 35 \times (0.6)^3 \times (0.4)^4 = 35 \times 0.216 \times 0.0256 = 0.193536, \\ P(X = 4) &= 35 \times (0.6)^4 \times (0.4)^3 = 35 \times 0.1296 \times 0.064 = 0.290304. \end{aligned}$$

$$P(2 \leq X \leq 4) = 0.077414 + 0.193536 + 0.290304 = 0.561254,$$

which rounds to 0.5613.

Answer. The probability of two, three, or four successful compilations is 0.5613.

Remarks

The binomial formula needs a fixed n before the trials begin. Problems 1.1 through 1.10 all have that fixed count. The mean alone, as in the cache example, is np, but a probability of an extreme count still requires the full sum. In the replica problem the mean is $9 \times 0.7 = 6.3$, and the acceptance region $X \geq 5$ sits below that mean, which is why the probability 0.9012 is high rather than close to one half.

2 The Poisson distribution

A Poisson count is used when events occur singly, at a constant average rate, and independently of one another. Typical computing examples are requests arriving at a gateway, faults in a large body of code, and crashes of a stable service. There is a rate, but there is no natural trial count n.

If $X \sim \text{Poisson}(\lambda)$, then

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots, \tag{3}$$

and

$$\text{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda, \qquad \text{SD}(X) = \sqrt{\lambda}. \tag{4}$$

Two useful facts are used below. Over disjoint intervals the rates add: a mean of λ per minute gives a mean of $t\lambda$ over t minutes. And the successive-term ratio is

$$\frac{P(X = k)}{P(X = k - 1)} = \frac{\lambda}{k}.$$

When λ is an integer, this ratio equals 1 at $k = \lambda$, so $P(X = \lambda) = P(X = \lambda - 1)$, and both of those values are modes.

Problem 2.1. API arrivals

Requests arrive at an API gateway as a Poisson process with mean 3 requests per second. For a one-second window, find the probability of exactly two requests and the probability of an empty window.

Solution. Let $X \sim \text{Poisson}(3)$. Appendix A gives $e^{-3} = 0.049787$.

$$P(X = 0) = 0.049787,$$
$$P(X = 2) = e^{-3}\frac{3^2}{2!} = 0.049787 \times 4.5 = 0.224042.$$

Rounded to four decimal places, these are 0.0498 and 0.2240.

Answer. $P(X = 2) = 0.2240$ and $P(X = 0) = 0.0498$.

Problem 2.2. Defects per thousand lines

Defects in a code base occur at a mean rate of 1.5 per thousand lines and are modelled as Poisson. For a module of one thousand lines, find the probability of no defect and the probability of at least one defect.

Solution. Let $X \sim \text{Poisson}(1.5)$. Appendix A gives $e^{-1.5} = 0.223130$.

$$P(X = 0) = 0.223130,$$

which rounds to 0.2231. The complement is

$$P(X \geq 1) = 1 - 0.223130 = 0.776870,$$

which rounds to 0.7769.

Answer. A clean thousand-line module has probability 0.2231. At least one defect has probability 0.7769.

Problem 2.3. Service crashes

A service crashes 0.4 times per month on average, and the monthly count is Poisson. Find the probability of a crash-free month and the probability of at least one crash.

Solution. Let $X \sim \text{Poisson}(0.4)$. Appendix A gives $e^{-0.4} = 0.670320$.

$$P(X = 0) = 0.670320,$$

which rounds to 0.6703. Then

$$P(X \geq 1) = 1 - 0.670320 = 0.329680,$$

which rounds to 0.3297.

Answer. The month is crash-free with probability 0.6703, and it has at least one crash with probability 0.3297.

Problem 2.4. Packet arrivals

Packets arrive at a network interface with mean 4 per millisecond. The count is Poisson. Find the probability of exactly four arrivals in a millisecond, and the probability of fewer than three.

Solution. Let $X \sim \text{Poisson}(4)$. Appendix A gives $e^{-4} = 0.018316$.

$$P(X = 4) = e^{-4}\frac{4^4}{24} = 0.018316 \times \frac{256}{24} = 0.018316 \times 10.666667 = 0.195371,$$

which rounds to 0.1954. Fewer than three arrivals is $X \leq 2$:

$$\begin{aligned} P(X = 0) &= 0.018316, \\ P(X = 1) &= 4 \times 0.018316 = 0.073264, \\ P(X = 2) &= \frac{4^2}{2} \times 0.018316 = 0.146528. \end{aligned}$$

$$P(X < 3) = 0.018316 + 0.073264 + 0.146528 = 0.238108,$$

which rounds to 0.2381.

Answer. $P(X = 4) = 0.1954$ and $P(X < 3) = 0.2381$.

Problem 2.5. Error lines in a log

Rare error lines occur at a mean rate of 2 per page, as a Poisson count.

1. Find the probability of at least three error lines on one page.
2. Using the additive property, find the probability of exactly four error lines in a three-page excerpt.

Solution.

1. Let $X \sim \text{Poisson}(2)$ on one page, with $e^{-2} = 0.135335$.

$$P(X \leq 2) = e^{-2}\left(1 + 2 + \frac{2^2}{2}\right) = 5 \times 0.135335 = 0.676675.$$

$$P(X \geq 3) = 1 - 0.676675 = 0.323325,$$

which rounds to 0.3233.

2. Over three pages the mean is $3 \times 2 = 6$. Let $Y \sim \text{Poisson}(6)$, with $e^{-6} = 0.002479$.

$$P(Y = 4) = 0.002479 \times \frac{6^4}{24} = 0.002479 \times \frac{1296}{24} = 0.002479 \times 54 = 0.133866,$$

which rounds to 0.1339.

Answer. (a) 0.3233. (b) 0.1339.

Problem 2.6. Deadlocks

Database deadlocks occur at a mean rate of 0.8 per hour and are modelled as Poisson. Find the probability of at most one deadlock in an hour.

Solution. Let $X \sim \text{Poisson}(0.8)$, with $e^{-0.8} = 0.449329$.

$$P(X = 0) = 0.449329,$$
$$P(X = 1) = 0.8 \times 0.449329 = 0.359463.$$

$$P(X \le 1) = 0.449329 + 0.359463 = 0.808792,$$

which rounds to 0.8088.

Answer. The probability of at most one deadlock in the hour is 0.8088.

Problem 2.7. Interrupt counts and the mode

A sensor records interrupts at a mean rate of 5 per minute. The count is Poisson. Show that $P(X = 5) = P(X = 4)$, compute the common value, and state the standard deviation.

Solution. The term ratio gives

$$\frac{P(X = 5)}{P(X = 4)} = \frac{\lambda}{5} = \frac{5}{5} = 1,$$

so the two probabilities are equal. Both are modes, because the ratio λ/k is greater than 1 for every $k < 5$ and less than 1 for every $k > 5$. Using $e^{-5} = 0.006738$,

$$P(X = 4) = 0.006738 \times \frac{5^4}{24} = 0.006738 \times \frac{625}{24} = 0.006738 \times 26.041667 = 0.175469,$$

which rounds to 0.1755. The same value is $P(X = 5)$. The standard deviation is

$$\text{SD}(X) = \sqrt{5} = 2.2361.$$

Answer. $P(X = 4) = P(X = 5) = 0.1755$. The modes are 4 and 5, and the standard deviation is 2.2361.

Problem 2.8. Print jobs

Jobs arrive at a printer queue as a Poisson process with mean 2.5 jobs per minute.

1. Find the probability of at least one job in a given minute.
2. Over a four-minute interval, find the expected number of jobs and the probability that the queue receives none.

Solution.

1. Let $X \sim \text{Poisson}(2.5)$. Appendix A gives $e^{-2.5} = 0.082085$, so

   $$P(X \ge 1) = 1 - 0.082085 = 0.917915,$$

   which rounds to 0.9179.

2. Rates add, so the four-minute count Y is Poisson with mean $4 \times 2.5 = 10$. Thus $\text{E}[Y] = 10$. Appendix A gives $e^{-10} = 0.000045$, and $P(Y = 0) = 0.000045$.

Answer. (a) 0.9179. (b) The expected number is 10, and the probability of no job is 0.000045.

Problem 2.9. Bad sectors

Reports of a bad sector occur at a mean rate of 0.2 per drive-day and are Poisson.

1. For one drive on one day, find the probability of no report and the probability of exactly one report.
2. Over 10 independent drive-days, find the probability of exactly three reports.

Solution.

1. Let $X \sim \text{Poisson}(0.2)$, with $e^{-0.2} = 0.818731$.

$$P(X = 0) = 0.818731,$$
$$P(X = 1) = 0.2 \times 0.818731 = 0.163746.$$

   These round to 0.8187 and 0.1637.

2. Ten drive-days have mean $10 \times 0.2 = 2$. Let $Y \sim \text{Poisson}(2)$, with $e^{-2} = 0.135335$.

$$P(Y = 3) = 0.135335 \times \frac{2^3}{6} = 0.135335 \times \frac{4}{3} = 0.180447,$$

   which rounds to 0.1804.

Answer. (a) 0.8187 and 0.1637. (b) 0.1804.

Problem 2.10. Cloud-function failures

Failures of a cloud function average 6 per hour and are modelled as Poisson. Find the mean, variance, and standard deviation, and find the probability of at most two failures in an hour.

Solution. For $X \sim \text{Poisson}(6)$,

$$\mathrm{E}[X] = 6, \qquad \mathrm{Var}(X) = 6, \qquad \mathrm{SD}(X) = \sqrt{6} = 2.4495.$$

With $e^{-6} = 0.002479$,

$$P(X = 0) = 0.002479,$$
$$P(X = 1) = 6 \times 0.002479 = 0.014874,$$
$$P(X = 2) = 18 \times 0.002479 = 0.044622.$$

$$P(X \le 2) = 0.002479 + 0.014874 + 0.044622 = 0.061975,$$

which rounds to 0.0620.

Answer. The mean and variance are both 6, the standard deviation is 2.4495, and $P(X \le 2) = 0.0620$. Seeing two or fewer failures in an hour is uncommon under this rate, about a 6.2% event.

Remarks

The Poisson mean and variance are the same number. That is a check, not a modelling assumption one can choose: if a count of arrivals has a variance far from its mean, a Poisson description is the wrong description. Adding the rate over a longer window, as with the three log pages and the four printer minutes, is legitimate only when the rate itself stays constant.

3 The normal distribution

Measurement errors, aggregated runtimes, and many other computing quantities pile up around a mean in a bell shape. The model for that shape is the normal distribution. If $X \sim N(\mu, \sigma^2)$, standardisation produces a standard normal variable Z:

$$Z = \frac{X - \mu}{\sigma}, \qquad P(X \le x) = \Phi\left(\frac{x - \mu}{\sigma}\right). \tag{5}$$

Appendix B gives $\Phi(z) = P(Z \le z)$. Symmetry supplies the negative side, $\Phi(-z) = 1 - \Phi(z)$. Because a normal random variable is continuous, $P(X = x) = 0$ and strict inequalities do not change a probability. The empirical rule is the special case $\mu \pm 2\sigma$: the table gives probability 0.9544, often remembered as "about 95%."

A percentile runs the standardisation backwards. The value exceeded by only the upper 5% of the population is

$$x = \mu + 1.645\,\sigma,$$

because $P(Z \leq 1.645) = 0.95$.

Problem 3.1. Build time

Build times, in minutes, follow $N(50, 10^2)$. Find the probability that a build finishes in under 60 minutes.

Solution.

$$z = \frac{60 - 50}{10} = 1.00.$$

Appendix B gives $\Phi(1.00) = 0.8413$.

Answer. $P(X < 60) = 0.8413$. About 84.13% of builds finish in under an hour.

Problem 3.2. Request latency

Latency, in milliseconds, follows $N(200, 25^2)$. Find the probability that a request takes more than 250 milliseconds.

Solution.

$$z = \frac{250 - 200}{25} = 2.00.$$

$$P(X > 250) = 1 - \Phi(2.00) = 1 - 0.9772 = 0.0228.$$

Answer. The probability of a latency above 250 ms is 0.0228.

Problem 3.3. Heap use

Heap use, in megabytes, follows $N(120, 15^2)$. Find the probability that usage lies strictly between 90 and 150 megabytes.

Solution. The endpoints are two standard deviations from the mean:

$$z_1 = \frac{90 - 120}{15} = -2.00, \qquad z_2 = \frac{150 - 120}{15} = 2.00.$$

$$P(90 < X < 150) = \Phi(2.00) - \Phi(-2.00) = 0.9772 - 0.0228 = 0.9544.$$

Answer. The probability is 0.9544.

Problem 3.4. Score percentile

Scores on a systems examination follow $N(70, 8^2)$. Find the 95th percentile of the score distribution.

Solution. The 95th percentile has standard score 1.645:

$$x = 70 + 1.645 \times 8 = 70 + 13.16 = 83.16.$$

Answer. The 95th percentile is 83.16. About 95% of candidates score 83.16 or below.

Problem 3.5. Frame size

Frame sizes, in bytes, follow $N(500, 40^2)$. Find the probability that a frame is smaller than 420 bytes.

Solution.

$$z = \frac{420 - 500}{40} = -2.00.$$

$$P(X < 420) = \Phi(-2.00) = 1 - 0.9772 = 0.0228.$$

Answer. The probability is 0.0228.

Problem 3.6. CPU temperature

Under a fixed workload, CPU temperature in degrees Celsius follows $N(65, 4^2)$. Find the probability that the temperature is between 61 and 73 degrees.

Solution.

$$z_1 = \frac{61 - 65}{4} = -1.00, \qquad z_2 = \frac{73 - 65}{4} = 2.00.$$

$$P(61 < X < 73) = \Phi(2.00) - \Phi(-1.00) = 0.9772 - (1 - 0.8413) = 0.9772 - 0.1587 = 0.8185.$$

Answer. The probability is 0.8185.

Problem 3.7. Download time

Download time, in seconds, follows $N(12, 2^2)$. Find the probability that a download takes more than 15 seconds.

Solution.

$$z = \frac{15 - 12}{2} = 1.50.$$

$$P(X > 15) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668.$$

Answer. The probability of a download longer than 15 seconds is 0.0668.

Problem 3.8. Benchmark score and the empirical rule

A benchmark score follows $N(600, 15^2)$. Find the probability that a score lies within two standard deviations of the mean, and compare the table value with the empirical rule.

Solution. Two standard deviations is $2 \times 15 = 30$, so the interval is $570 < X < 630$.

$$z = \pm\frac{30}{15} = \pm 2.00.$$

$$P(570 < X < 630) = \Phi(2.00) - \Phi(-2.00) = 0.9772 - 0.0228 = 0.9544.$$

The empirical rule says "about 95%." The table makes that statement precise: 95.44%.

Answer. The probability is 0.9544.

Problem 3.9. Slow-response threshold

Response time, in milliseconds, follows $N(80, 10^2)$. Find the response time that is exceeded by only the slowest 5% of requests.

Solution. The upper 5% point of the standard normal is 1.645, so

$$x = 80 + 1.645 \times 10 = 96.45.$$

Answer. The slowest 5% of requests take longer than 96.45 ms.

Problem 3.10. Fail and distinction rates

Course totals follow $N(65, 12^2)$. A total below 50 fails. A total of 80 or above is a distinction. Find both probabilities.

Solution.

$$z_{\text{fail}} = \frac{50 - 65}{12} = -1.25, \qquad z_{\text{distinction}} = \frac{80 - 65}{12} = 1.25.$$

Appendix B gives $\Phi(1.25) = 0.8944$, so $\Phi(-1.25) = 1 - 0.8944 = 0.1056$.

$$P(X < 50) = 0.1056,$$
$$P(X \geq 80) = 1 - \Phi(1.25) = 1 - 0.8944 = 0.1056.$$

The two probabilities agree because 50 and 80 are the same distance, 15 marks, from the mean 65.

Answer. The failure probability and the distinction probability are both 0.1056.

Remarks

Every probability in this section was a table lookup after one subtraction and one division. That is the practical point of standardisation. The same table does not, by itself, say that build times or examination scores are normal. It says what the probabilities are if that model has already been accepted. For a count, rather than a measurement, the passage from a binomial model to a normal model needs the checks in Section 5.

4 The Poisson approximation to the binomial

The binomial probability $\binom{n}{k}p^k(1-p)^{n-k}$ is awkward when n is large. When successes are also rare, the Poisson formula is a close substitute. The working rule in this paper is

use Poisson(λ) with $\lambda = np$ when $n \geq 50$ and $p \leq 0.1$.

Many textbooks use the stricter pair $n \geq 20$ and $p \leq 0.05$. Every approximation below satisfies $n \geq 50$ and $p \leq 0.05$, except the comparison exercise, which is computed both ways because the exact binomial happens to simplify.

The approximation replaces

$$P(X = k) \approx e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \lambda = np.$$

It does not change the mean: both models have mean np. The binomial variance $np(1-p)$ is slightly smaller than λ, and that gap is the main source of the error. For a first look at the size of the error, three exact values computed outside the hand arithmetic are:

Case	Exact binomial	Poisson	Absolute error
$n = 100,\ p = 0.01,\ X = 1$	0.3697	0.3679	0.0018
$n = 100,\ p = 0.02,\ X = 3$	0.1823	0.1804	0.0018
$n = 200,\ p = 0.01,\ X = 0$	0.1340	0.1353	0.0014

The hand solutions compute the Poisson column. Problem 4.9 also computes the exact binomial, because $\binom{100}{1}(0.01)(0.99)^{99}$ collapses to a single power.

Problem 4.1. Node failures

In a cluster of 100 nodes, each node fails during a week with probability 0.02, independently. Approximate the probability that exactly three nodes fail.

Solution. Check the rule: $n = 100 \geq 50$ and $p = 0.02 \leq 0.1$. The mean is

$$\lambda = 100 \times 0.02 = 2.$$

With $e^{-2} = 0.135335$,

$$P(X = 3) \approx 0.135335 \times \frac{2^3}{6} = 0.135335 \times \frac{4}{3} = 0.180447,$$

which rounds to 0.1804. The exact binomial value is 0.1823, so the approximation is 0.0018 low.

Answer. The Poisson approximation is 0.1804.

Problem 4.2. Packet corruption

Of 200 packets, each is corrupted with probability 0.01, independently. Approximate the probability that none are corrupted.

Solution. Here $n = 200$, $p = 0.01$, and $\lambda = 2$. The conditions hold.

$$P(X = 0) \approx e^{-2} = 0.135335,$$

which rounds to 0.1353.

Answer. The approximate probability of a clean batch is 0.1353.

Problem 4.3. Drive failures

Fifty drives are tested. Each fails the test with probability 0.04, independently. Approximate the probability of at most one failure.

Solution. The conditions hold, and $\lambda = 50 \times 0.04 = 2$. Using $e^{-2} = 0.135335$,

$$P(X = 0) \approx 0.135335,$$
$$P(X = 1) \approx 2 \times 0.135335 = 0.270670.$$

$$P(X \le 1) \approx 0.135335 + 0.270670 = 0.406005,$$

which rounds to 0.4060.

Answer. The approximate probability of at most one failure is 0.4060.

Problem 4.4. Transaction deadlocks

Eighty transactions are submitted. Each meets a deadlock with probability 0.05, independently. Approximate the probability of exactly four deadlocks.

Solution. Here $n = 80 \ge 50$, $p = 0.05 \le 0.1$, and $\lambda = 80 \times 0.05 = 4$. With $e^{-4} = 0.018316$,

$$P(X = 4) \approx 0.018316 \times \frac{4^4}{24} = 0.018316 \times \frac{256}{24}.$$

Since $256/24 = 10.666667$,

$$0.018316 \times 10.666667 = 0.195371,$$

which rounds to 0.1954.

Answer. The approximate probability is 0.1954.

Problem 4.5. Packet loss

A thousand packets are sent. Each is lost with probability 0.003, independently. Approximate the probability that at least one packet is lost.

Solution. The conditions hold, and $\lambda = 1000 \times 0.003 = 3$. The empty count is the cheap term:

$$P(X = 0) \approx e^{-3} = 0.049787.$$

$$P(X \ge 1) \approx 1 - 0.049787 = 0.950213,$$

which rounds to 0.9502.

Answer. The approximate probability of at least one loss is 0.9502.

Problem 4.6. Typos in generated pages

A documentation build produces 120 pages. Each page contains a typo with probability 0.025, independently. Approximate the probability that exactly two pages contain a typo.

Solution. Here $\lambda = 120 \times 0.025 = 3$, and the conditions hold. With $e^{-3} = 0.049787$,

$$P(X = 2) \approx 0.049787 \times \frac{3^2}{2} = 0.049787 \times 4.5 = 0.224042,$$

which rounds to 0.2240.

Answer. The approximate probability is 0.2240.

Problem 4.7. Unreadable sectors

Four hundred sectors are read. Each is unreadable with probability 0.005, independently. Approximate the probability that more than two sectors are unreadable.

Solution. Here $\lambda = 400 \times 0.005 = 2$. More than two means the complement of 0, 1, and 2:

$$P(X \le 2) \approx e^{-2}\left(1 + 2 + \frac{2^2}{2}\right) = 5 \times 0.135335 = 0.676675.$$

$$P(X > 2) \approx 1 - 0.676675 = 0.323325,$$

which rounds to 0.3233.

Answer. The approximate probability of more than two unreadable sectors is 0.3233.

Problem 4.8. Query timeouts

One hundred and fifty queries are issued. Each times out with probability 0.02, independently. Approximate the probability of exactly five timeouts.

Solution. Here $\lambda = 150 \times 0.02 = 3$. With $e^{-3} = 0.049787$ and $3^5 = 243$,

$$P(X = 5) \approx 0.049787 \times \frac{243}{120} = 0.049787 \times 2.025 = 0.100819,$$

which rounds to 0.1008.

Answer. The approximate probability is 0.1008.

Problem 4.9. Exact binomial against the approximation

One hundred independent trials each succeed with probability 0.01. Compute $P(X = 1)$ from the binomial formula and from the Poisson approximation. Report both probabilities and the absolute error.

Solution. The exact probability collapses:

$$P(X = 1) = \binom{100}{1}(0.01)^1(0.99)^{99} = (0.99)^{99}.$$

The power evaluates to 0.36972964, which rounds to 0.3697.

For the approximation, $\lambda = 100 \times 0.01 = 1$ and

$$P(X = 1) \approx e^{-1}\frac{1^1}{1!} = 0.367879,$$

which rounds to 0.3679. The absolute error of the unrounded values is 0.0018. Both $n \ge 50$ and $p \le 0.1$ hold, and an error of about two thousandths is typical of this range rather than a warning that the rule has failed.

Answer. Exact probability 0.3697; Poisson approximation 0.3679; absolute error 0.0018.

Problem 4.10. Chip defects

A batch contains 500 chips. Each chip is defective with probability 0.004, independently. Using the Poisson approximation, find the mean, the variance of the approximating distribution, and the probability of at least four defectives.

Solution. The conditions hold, and $\lambda = 500 \times 0.004 = 2$. The approximating Poisson distribution has mean 2 and variance 2. The binomial variance would be $np(1-p) = 2 \times 0.996 = 1.992$, almost the same number. For the upper tail, with $e^{-2} = 0.135335$,

$$P(X = 0) \approx 0.135335,$$

$$P(X = 1) \approx 0.270670,$$

$$P(X = 2) \approx 0.270670,$$

$$P(X = 3) \approx 0.135335 \times \frac{4}{3} = 0.180447.$$

$$P(X \le 3) \approx 0.135335 + 0.270670 + 0.270670 + 0.180447 = 0.857122.$$

$$P(X \ge 4) \approx 1 - 0.857122 = 0.142878,$$

which rounds to 0.1429.

Answer. The approximating mean and variance are both 2, and $P(X \ge 4) \approx 0.1429$.

Remarks

The approximation is a computational convenience, not a different physical story. Once $\lambda = np$ is in hand, every technique from Section 2 applies, including complements and the addition of rates. If p is not small, the variance gap $np - np(1-p) = np^2$ stops being negligible and the normal approximation of the next section is the one to consider.

5 The normal approximation to the binomial

When both the expected number of successes and the expected number of failures are reasonably large, the binomial histogram looks like a normal bell. The rule used here is

$$X \sim \mathrm{Bin}(n,p) \text{ may be approximated by } N(\mu,\sigma^2) \text{ when } np \geq 5 \text{ and } n(1-p) \geq 5,$$

with

$$\mu = np, \qquad \sigma = \sqrt{np(1-p)}.$$

A count is discrete and the normal curve is continuous, so each integer k is given the unit interval from $k - 0.5$ to $k + 0.5$. This continuity correction is used in every solution below.

Binomial probability	Normal approximation
$P(X \leq k)$	$\Phi\big((k + 0.5 - \mu)/\sigma\big)$
$P(X \geq k)$	$1 - \Phi\big((k - 0.5 - \mu)/\sigma\big)$
$P(X = k)$	$\Phi\big((k + 0.5 - \mu)/\sigma\big) - \Phi\big((k - 0.5 - \mu)/\sigma\big)$
$P(a \leq X \leq b)$	$\Phi\big((b + 0.5 - \mu)/\sigma\big) - \Phi\big((a - 0.5 - \mu)/\sigma\big)$

The parameters in this section are chosen so that σ is 5 or 10. The arithmetic of the correction is then visible. The same steps apply when σ is not an integer.

Problem 5.1. Load balancer

One hundred independent requests are each sent to server A with probability 0.5. Approximate the probability that server A receives between 45 and 55 requests, inclusive.

Solution. Here $np = 50$ and $n(1-p) = 50$, so the normal approximation is appropriate.

$$\mu = 50, \qquad \sigma = \sqrt{100 \times 0.5 \times 0.5} = \sqrt{25} = 5.$$

The corrected interval runs from 44.5 to 55.5:

$$z_1 = \frac{44.5 - 50}{5} = -1.10, \qquad z_2 = \frac{55.5 - 50}{5} = 1.10.$$

$$P(45 \leq X \leq 55) \approx \Phi(1.10) - \Phi(-1.10) = 0.8643 - 0.1357 = 0.7286.$$

Answer. The approximate probability is 0.7286.

Problem 5.2. Cache hits, upper tail

Of 625 independent lookups, each is a cache hit with probability 0.2. Approximate the probability of at least 140 hits.

Solution. Check $np = 625 \times 0.2 = 125 \geq 5$ and $n(1-p) = 500 \geq 5$. Then

$$\sigma = \sqrt{625 \times 0.2 \times 0.8} = \sqrt{100} = 10.$$

The event $X \geq 140$ starts at the corrected boundary 139.5:

$$z = \frac{139.5 - 125}{10} = 1.45.$$

$$P(X \geq 140) \approx 1 - \Phi(1.45) = 1 - 0.9265 = 0.0735.$$

The count 625 was chosen so that the variance is a square. Nothing else in the method depends on that choice.

Answer. The approximate probability is 0.0735.

Problem 5.3. Balanced assignments, lower tail

Four hundred independent requests are each assigned to a canary build with probability 0.5. Approximate the probability that the canary receives at most 180 requests.

Solution. Here $\mu = 200$ and $\sigma = \sqrt{400 \times 0.5 \times 0.5} = 10$. Both np and $n(1-p)$ equal 200, so the approximation is valid. For $X \le 180$ the corrected boundary is 180.5:

$$z = \frac{180.5 - 200}{10} = -1.95.$$

$$P(X \le 180) \approx \Phi(-1.95) = 1 - \Phi(1.95) = 1 - 0.9744 = 0.0256.$$

Answer. The approximate probability is 0.0256.

Problem 5.4. A single central count

One hundred fair assignments are made, each to pool A with probability 0.5. Approximate the probability that pool A receives exactly 50 assignments.

Solution. Here $\mu = 50$ and $\sigma = 5$. A single integer occupies (49.5, 50.5):

$$z_1 = \frac{49.5 - 50}{5} = -0.10, \qquad z_2 = \frac{50.5 - 50}{5} = 0.10.$$

$$P(X = 50) \approx \Phi(0.10) - \Phi(-0.10) = 0.5398 - 0.4602 = 0.0796.$$

A normal approximation of one integer is only a local area of width 1. The correction is the whole calculation: without it the continuous model would give this probability as zero.

Answer. The approximate probability is 0.0796.

Problem 5.5. Successful health checks

Of 625 independent health checks, each succeeds with probability 0.8. Approximate the probability that the number of successes is between 490 and 510, inclusive.

Solution. Here $\mu = 625 \times 0.8 = 500$ and

$$\sigma = \sqrt{625 \times 0.8 \times 0.2} = \sqrt{100} = 10.$$

Both expected counts, 500 and 125, exceed 5. The corrected endpoints are 489.5 and 510.5:

$$z_1 = \frac{489.5 - 500}{10} = -1.05, \qquad z_2 = \frac{510.5 - 500}{10} = 1.05.$$

$$P(490 \le X \le 510) \approx \Phi(1.05) - \Phi(-1.05) = 0.8531 - 0.1469 = 0.7062.$$

Answer. The approximate probability is 0.7062.

Problem 5.6. Overloaded server

Four hundred requests are each routed to a server with probability 0.5. Approximate the probability that the server receives more than 220 requests.

Solution. More than 220 is $X \ge 221$, because a count is an integer. Then $\mu = 200$, $\sigma = 10$, and the corrected boundary is 220.5:

$$z = \frac{220.5 - 200}{10} = 2.05.$$

$$P(X > 220) \approx 1 - \Phi(2.05) = 1 - 0.9798 = 0.0202.$$

Answer. The approximate probability is 0.0202.

Problem 5.7. Too few assignments

Using the same 100 fair assignments as in Problem 5.4, approximate the probability that pool A receives at most 39.

Solution. The mean and standard deviation are again $\mu = 50$ and $\sigma = 5$. For $X \le 39$,

$$z = \frac{39.5 - 50}{5} = -2.10.$$

$$P(X \le 39) \approx \Phi(-2.10) = 1 - 0.9821 = 0.0179.$$

Answer. The approximate probability is 0.0179. A count this far below the mean is unusual under a fair assignment.

Problem 5.8. When the normal approximation is not valid

Forty overnight jobs each fail with probability 0.05, independently. A first draft approximates $P(X = 2)$ by a normal curve. Check the condition. If the normal approximation is not justified, compute a Poisson approximation instead.

Solution. Here $n = 40$ and $p = 0.05$, so

$$np = 2, \qquad n(1-p) = 38.$$

The expected number of failures is 2, which is below 5. The normal approximation is not justified: the histogram is piled against zero and is not a symmetric bell.

The Poisson approximation is the matching rare-event tool. Its usual sample-size rule $n \ge 50$ is not quite met, but $p = 0.05$ and $\lambda = np = 2$ still make it the appropriate classroom approximation. With $e^{-2} = 0.135335$,

$$P(X = 2) \approx 0.135335 \times \frac{2^2}{2} = 0.270670,$$

which rounds to 0.2707.

Answer. The normal approximation is not valid because $np = 2 < 5$. The Poisson approximation is $P(X = 2) \approx 0.2707$.

Problem 5.9. A central band of cache hits

Of 625 lookups, each hits with probability 0.2, independently. Approximate the probability that the number of hits is between 110 and 130, inclusive.

Solution. From Problem 5.2, $\mu = 125$ and $\sigma = 10$, and the normal approximation is valid. The corrected endpoints of $110 \le X \le 130$ are 109.5 and 130.5:

$$z_1 = \frac{109.5 - 125}{10} = -1.55, \qquad z_2 = \frac{130.5 - 125}{10} = 0.55.$$

$$P(110 \le X \le 130) \approx \Phi(0.55) - \Phi(-1.55) = 0.7088 - (1 - 0.9394) = 0.7088 - 0.0606 = 0.6482.$$

Answer. The approximate probability is 0.6482.

Problem 5.10. A rare alert

Four hundred requests are each sent to server A with probability 0.5. An alert fires if server A receives at least 230 requests. Approximate the probability that the alert fires on a fair assignment.

Solution. Here $\mu = 200$ and $\sigma = 10$. For $X \ge 230$ the corrected boundary is 229.5:

$$z = \frac{229.5 - 200}{10} = 2.95.$$

$$P(X \ge 230) \approx 1 - \Phi(2.95) = 1 - 0.9984 = 0.0016.$$

Answer. Under a fair assignment the alert has approximate probability 0.0016. It is a rare event, not a routine fluctuation of a few requests.

Remarks

The continuity correction moves an endpoint by half a count. That shift is 0.1 of a standard deviation when $\sigma = 10$, and 0.5 of a standard deviation when $\sigma = 1$. It therefore matters most for a single integer or a short interval, which is why Problem 5.4 is nothing but the correction. The condition $np \geq 5$ and $n(1-p) \geq 5$ is a gate, not a decoration: Problem 5.8 fails it, and using the bell curve there would describe a shape the count does not have.

6 The negative binomial distribution

A negative binomial random variable is a waiting time. Fix a success probability p and a required number of successes r. Let X be the number of independent trials needed to obtain those r successes. Then

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, r+2, \ldots. \tag{6}$$

The mean and variance are

$$\mathrm{E}[X] = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}. \tag{7}$$

The binomial coefficient counts the ways to place the first $r-1$ successes among the first $k-1$ trials, so that trial k is the one that completes the quota. The last trial is a success; it is not one of the free positions.

Some courses instead let Y be the number of failures before the rth success. Then $X = Y + r$,

$$P(Y = y) = \binom{y+r-1}{r-1} p^r (1-p)^y, \qquad y = 0, 1, 2, \ldots, \tag{8}$$

and $\mathrm{E}[Y] = r(1-p)/p$. Problem 6.8 uses this second form and translates it back to X. Everywhere else, X is the number of trials.

When $r = 1$, the negative binomial distribution is the geometric distribution of the next section.

Problem 6.1. Flaky integration test

An integration test passes with probability 0.25, independently from run to run. A release script reruns it until it has passed three times. Find the probability that the third pass occurs on the fifth run.

Solution. Here $r = 3$, $p = 0.25$, and $k = 5$. The first four runs contain exactly two passes, and the fifth run passes:

$$P(X = 5) = \binom{4}{2} (0.25)^3 (0.75)^2.$$

$\binom{4}{2} = 6$, $(0.25)^3 = 0.015625$, and $(0.75)^2 = 0.5625$.

$$P(X = 5) = 6 \times 0.015625 \times 0.5625 = 0.09375 \times 0.5625 = 0.052734,$$

which rounds to 0.0527.

Answer. The probability is 0.0527.

Problem 6.2. Two heads

A fair simulator produces heads with probability 0.5. It is run until the second head. Find the probability that this takes exactly four runs.

Solution. Here $r = 2$, $p = 0.5$, and $k = 4$.

$$P(X = 4) = \binom{3}{1} (0.5)^2 (0.5)^2 = 3 \times \frac{1}{16} = \frac{3}{16} = 0.1875.$$

The first three runs contain exactly one head, and the fourth run is a head.

Answer. The probability is 0.1875.

Problem 6.3. Fuzzer stopping rule

A fuzzer finds a crashing input with probability 0.2 on each independent trial. It stops after the fourth crash. Find the expected number of trials, the variance, and the standard deviation.

Solution. Here $r = 4$ and $p = 0.2$, so $1 - p = 0.8$.

$$\begin{aligned} \mathrm{E}[X] &= \frac{4}{0.2} = 20, \\ \mathrm{Var}(X) &= \frac{4 \times 0.8}{(0.2)^2} = \frac{3.2}{0.04} = 80, \\ \mathrm{SD}(X) &= \sqrt{80} = 8.9443. \end{aligned}$$

Answer. The expected number of trials is 20, the variance is 80, and the standard deviation is 8.9443.

Problem 6.4. Second crashing input

A random input crashes a parser with probability 0.1. Inputs are drawn independently until the second crash. Find the expected number of inputs and the probability that the second crash occurs on the fifth input.

Solution. Here $r = 2$ and $p = 0.1$.

$$\mathrm{E}[X] = \frac{2}{0.1} = 20.$$

For $k = 5$,

$$P(X = 5) = \binom{4}{1}(0.1)^2(0.9)^3 = 4 \times 0.01 \times 0.729 = 0.02916,$$

which rounds to 0.0292.

Answer. The expected number of inputs is 20, and $P(X = 5) = 0.0292$.

Problem 6.5. Five acknowledgements

A packet transmission is acknowledged with probability 0.8, independently. A sender waits until it has collected five acknowledgements. Find the probability that this happens in exactly five transmissions, and find the expected number of transmissions.

Solution. Exactly five transmissions means that every transmission is acknowledged:

$$P(X = 5) = (0.8)^5 = 0.32768,$$

which rounds to 0.3277. There is no binomial coefficient other than 1, because $\binom{4}{4} = 1$: every one of the first four trials must already be a success. The expectation is

$$\mathrm{E}[X] = \frac{5}{0.8} = 6.25.$$

Answer. $P(X = 5) = 0.3277$ and the expected number of transmissions is 6.25.

Problem 6.6. At most four trials

Each trial succeeds with probability 0.4. Find the probability that the third success occurs on trial 3 or trial 4, that is, that at most four trials are required.

Solution. Let $r = 3$ and $p = 0.4$. The support starts at $k = 3$, so

$$P(X \le 4) = P(X = 3) + P(X = 4).$$

$$P(X = 3) = (0.4)^3 = 0.064,$$
$$P(X = 4) = \binom{3}{2}(0.4)^3(0.6) = 3 \times 0.064 \times 0.6 = 0.1152.$$

$$P(X \le 4) = 0.064 + 0.1152 = 0.1792.$$

Answer. The probability that the third success arrives by the fourth trial is 0.1792.

Problem 6.7. Second relevant record

A probe of an index hits a relevant record with probability 0.3. Probes are independent. Find the probability that the second relevant record occurs on the sixth probe.

Solution. Here $r = 2$, $p = 0.3$, and $k = 6$.

$$P(X = 6) = \binom{5}{1}(0.3)^2(0.7)^4.$$

$(0.7)^2 = 0.49$ and $(0.7)^4 = 0.2401$, so

$$P(X = 6) = 5 \times 0.09 \times 0.2401 = 0.45 \times 0.2401 = 0.108045,$$

which rounds to 0.1080.

Answer. The probability is 0.1080.

Problem 6.8. Failures before the third success

A connection attempt succeeds with probability 0.5. Let Y be the number of failures before the third success, and let X be the number of attempts needed to obtain the third success.

1. Find $P(Y = 2)$ from the failures formula.
2. Find $\mathrm{E}[Y]$.
3. Translate the event $Y = 2$ into a statement about X, and check it with the trials formula.

Solution.

1. With $r = 3$, $p = 0.5$, and $y = 2$,
$$P(Y = 2) = \binom{2+3-1}{2}(0.5)^3(0.5)^2 = \binom{4}{2}\frac{1}{32} = \frac{6}{32} = 0.1875.$$
2. $$\mathrm{E}[Y] = \frac{r(1-p)}{p} = \frac{3 \times 0.5}{0.5} = 3.$$
3. Two failures before the third success means five attempts in all, so $Y = 2$ is the same event as $X = 5$. The trials formula gives
$$P(X = 5) = \binom{4}{2}(0.5)^3(0.5)^2 = 0.1875,$$
the same value. Also $\mathrm{E}[X] = \mathrm{E}[Y] + r = 3 + 3 = 6$, which agrees with $r/p = 6$.

Answer. (a) 0.1875. (b) 3. (c) $Y = 2$ is the same event as $X = 5$, and both formulas give 0.1875.

Problem 6.9. More than four trials

Each trial succeeds with probability 0.6. Find the probability that more than four trials are needed to obtain the second success.

Solution. Let $r = 2$ and $p = 0.6$. Then

$$P(X > 4) = 1 - P(X = 2) - P(X = 3) - P(X = 4).$$

$$P(X = 2) = (0.6)^2 = 0.36,$$
$$P(X = 3) = \binom{2}{1}(0.6)^2(0.4) = 2 \times 0.36 \times 0.4 = 0.288,$$
$$P(X = 4) = \binom{3}{1}(0.6)^2(0.4)^2 = 3 \times 0.36 \times 0.16 = 0.1728.$$

$$P(X \le 4) = 0.36 + 0.288 + 0.1728 = 0.8208,$$
$$P(X > 4) = 1 - 0.8208 = 0.1792.$$

Answer. The probability of needing more than four trials is 0.1792.

Problem 6.10. Scaling with the number of successes

Compare a geometric wait for one success with a negative binomial wait for four successes, both with success probability 0.25. Compute the mean and variance of each, and state the relationship between them.

Solution. For $r = 1$,

$$\mathrm{E}[X_1] = \frac{1}{0.25} = 4, \qquad \mathrm{Var}(X_1) = \frac{0.75}{(0.25)^2} = \frac{0.75}{0.0625} = 12.$$

For $r = 4$,

$$\mathrm{E}[X_4] = \frac{4}{0.25} = 16, \qquad \mathrm{Var}(X_4) = \frac{4 \times 0.75}{0.0625} = 48.$$

Both the mean and the variance are multiplied by 4. That is the general fact $\mathrm{E}[X_r] = r\,\mathrm{E}[X_1]$ and $\mathrm{Var}(X_r) = r\,\mathrm{Var}(X_1)$: the wait for r successes is the sum of r independent geometric waits.

Answer. One success: mean 4, variance 12. Four successes: mean 16, variance 48. The four-success wait scales both moments by 4.

Remarks

The support starts at r, not at 0. A probability assigned to $k = 2$ when three successes are required is a signal that the wrong formula, or the failures convention, has been used. The two conventions differ by exactly r, as Problem 6.8 checks on a single event.

7 The geometric distribution

The geometric distribution is the negative binomial distribution with $r = 1$: the number of independent trials needed for the first success. If each trial succeeds with probability p, then

$$P(X = k) = (1 - p)^{k-1}p, \qquad k = 1, 2, 3, \ldots, \tag{9}$$

$$\mathrm{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1 - p}{p^2}, \qquad P(X > k) = (1 - p)^k. \tag{10}$$

The survival formula is the one used for a budget. The probability of needing more than k trials is the probability that the first k trials all fail. The cumulative probability follows at once:

$$P(X \le k) = 1 - (1 - p)^k.$$

The distribution is memoryless. For positive integers s and t,

$$P(X > s + t \mid X > s) = P(X > t). \tag{11}$$

Failures already observed do not reduce the number of future trials still expected. A stateless retry loop has exactly this behaviour.

Some authors let the geometric variable count failures before the first success, so that its values start at 0. This paper counts trials and starts at 1. The two versions differ by one, just as X and Y differed by r in the previous section.

Problem 7.1. Connection retry

A client retries a connection that succeeds with probability 0.2 on each independent attempt. Find the probability that the first success occurs on the fourth attempt. Also find the mean, variance, and standard deviation of the number of attempts.

Solution.

$$P(X = 4) = (0.8)^3(0.2) = 0.512 \times 0.2 = 0.1024.$$

$$\begin{aligned} \mathrm{E}[X] &= \frac{1}{0.2} = 5, \\ \mathrm{Var}(X) &= \frac{0.8}{(0.2)^2} = \frac{0.8}{0.04} = 20, \\ \mathrm{SD}(X) &= \sqrt{20} = 4.4721. \end{aligned}$$

Answer. $P(X = 4) = 0.1024$. The mean is 5, the variance is 20, and the standard deviation is 4.4721.

Problem 7.2. Success within three trials

A trial succeeds with probability 0.5. Find the probability that the first success occurs within the first three trials.

Solution.

$$P(X \le 3) = 1 - (0.5)^3 = 1 - 0.125 = 0.875.$$

The same result comes from adding the point masses:

$$P(X = 1) + P(X = 2) + P(X = 3) = 0.5 + 0.25 + 0.125 = 0.875.$$

Answer. The probability is 0.875.

Problem 7.3. A long run of failures

A rare check succeeds with probability 0.1. Find the probability that the first success has not occurred by the fifth trial, and state what that event means in terms of the five individual trials.

Solution.

$$P(X > 5) = (0.9)^5 = 0.59049,$$

which rounds to 0.5905. The event $X > 5$ means that all of the first five trials failed. It does not mean that the fifth trial failed and the earlier ones are unknown; every one of them failed.

Answer. The probability is 0.5905.

Problem 7.4. Success on an odd trial

A trial succeeds with probability 0.3. Find the probability that the first success occurs on trial 1, 3, 5, or any later odd trial.

Solution. Let $q = 0.7$. The required probability is the infinite sum

$$P(X \text{ is odd}) = \sum_{j=0}^{\infty} q^{2j} p = \frac{p}{1 - q^2},$$

a geometric series with first term p and ratio q^2. Substituting the values,

$$P = \frac{0.3}{1-(0.7)^2} = \frac{0.3}{1-0.49} = \frac{0.3}{0.51} = 0.588235,$$

which rounds to 0.5882. The probability is a little above one half because the very first trial, which is odd, already carries probability 0.3, and the remaining odd trials add to that lead.

Answer. The probability is 0.5882.

Problem 7.5. Memoryless retries

A connection attempt succeeds with probability 0.25. The first three attempts have already failed. Find the probability that the next two attempts also fail. Explain why the three failures already spent do not appear in the answer.

Solution. Memorylessness says

$$P(X > 5 \mid X > 3) = P(X > 2) = (0.75)^2 = 0.5625.$$

The condition $X > 3$ is the information that the first three attempts failed. The event $X > 5$ says that attempt 4 and attempt 5 fail as well. Because a geometric wait does not age, the future two trials have the same failure probability they would have had at the start. The three spent failures change the clock, not the distribution of the remaining wait.

Answer. The conditional probability is 0.5625.

Problem 7.6. First checksum failure

Packets fail a checksum with probability 0.05, independently. Inspection stops at the first failure. Find the expected number of packets inspected, the standard deviation, and the probability that the first packet is the one that fails.

Solution.

$$\begin{aligned} \mathrm{E}[X] &= \frac{1}{0.05} = 20, \\ \mathrm{Var}(X) &= \frac{0.95}{(0.05)^2} = \frac{0.95}{0.0025} = 380, \\ \mathrm{SD}(X) &= \sqrt{380} = 19.4936. \end{aligned}$$

The first packet fails with probability $P(X = 1) = p = 0.05$. The standard deviation is almost as large as the mean, which is typical when p is small: most inspections are short of 20 packets, and a long right tail pulls the mean up.

Answer. The mean is 20, the standard deviation is 19.4936, and $P(X = 1) = 0.05$.

Problem 7.7. Two links

Link A succeeds on a given attempt with probability 0.4. Link B succeeds with probability 0.8. Each link is retried until its first success. Compare the expected number of attempts and the standard deviations.

Solution. For link A,

$$\mathrm{E}[X_A] = \frac{1}{0.4} = 2.5, \qquad \mathrm{Var}(X_A) = \frac{0.6}{0.16} = 3.75, \qquad \mathrm{SD}(X_A) = \sqrt{3.75} = 1.9365.$$

For link B,

$$\mathrm{E}[X_B] = \frac{1}{0.8} = 1.25, \qquad \mathrm{Var}(X_B) = \frac{0.2}{0.64} = 0.3125, \qquad \mathrm{SD}(X_B) = \sqrt{0.3125} = 0.5590.$$

Link B needs half as many attempts on average, because its success probability is twice as large. Its standard deviation is much smaller as well: a success probability of 0.8 leaves little room for a long string of failures.

Answer. Link A: mean 2.5, standard deviation 1.9365. Link B: mean 1.25, standard deviation 0.5590.

Problem 7.8. Open addressing

In a simplified model of open addressing, each independent probe finds an empty slot with probability 0.6. Let X be the probe number of the first empty slot. Find $P(X = 3)$, $P(X \geq 3)$, and $\mathrm{E}[X]$.

Solution.

$$P(X = 3) = (0.4)^2(0.6) = 0.16 \times 0.6 = 0.096,$$
$$P(X \geq 3) = P(X > 2) = (0.4)^2 = 0.16,$$
$$\mathrm{E}[X] = \frac{1}{0.6} = 1.6667.$$

The event $X \geq 3$ is the event that the first two probes were occupied. The third probe is not required to be empty for that statement; it may be empty or occupied.

Answer. $P(X = 3) = 0.0960$, $P(X \geq 3) = 0.1600$, and the expected number of probes is 1.6667.

Problem 7.9. Trials needed for 95 percent confidence of a success

A trial succeeds with probability 0.3. Find the smallest integer k such that $P(X \leq k) \geq 0.95$. This is the smallest budget that gives at least a 95% chance of having seen the first success.

Solution. The cumulative formula rearranges as

$$1 - (0.7)^k \geq 0.95 \qquad \Longleftrightarrow \qquad (0.7)^k \leq 0.05.$$

Taking natural logarithms, and reversing the inequality because $\ln 0.7 < 0$,

$$k \geq \frac{\ln 0.05}{\ln 0.7} = \frac{-2.995732}{-0.356675} = 8.399.$$

The smallest integer k is therefore 9. Direct powers confirm the boundary:

$$(0.7)^8 = 0.057648, \qquad P(X \leq 8) = 1 - 0.057648 = 0.942352,$$
$$(0.7)^9 = 0.040354, \qquad P(X \leq 9) = 1 - 0.040354 = 0.959646.$$

So $k = 8$ falls short of 0.95, and $k = 9$ clears it. Rounded, the two cumulative probabilities are 0.9424 and 0.9596.

Answer. The smallest such budget is $k = 9$ trials.

Problem 7.10. Retry budget in milliseconds

Each attempt costs 4 milliseconds, and an attempt succeeds with probability 0.25. Let $T = 4X$ be the time until the first success.

1. Find $\mathrm{E}[T]$.
2. Find the probability that the time strictly exceeds 16 milliseconds.

Solution.

1. $\mathrm{E}[X] = 1/0.25 = 4$, so $\mathrm{E}[T] = 4 \times 4 = 16$ milliseconds.
2. The possible times are $4, 8, 12, 16, 20, \ldots$ milliseconds. Strictly more than 16 means $T \geq 20$, hence $X \geq 5$, or equivalently $X > 4$.
$$P(X > 4) = (0.75)^4.$$
Since $(0.75)^2 = 0.5625$,
$$(0.75)^4 = 0.5625^2 = 0.316406,$$
which rounds to 0.3164.

The mean time is 16 ms, but the probability of exceeding that mean is only 0.3164, not 0.5. The geometric distribution is skewed to the right: many runs finish on the first few attempts, and a minority of long runs pull the average up to 16.

Answer. (a) 16 ms. (b) 0.3164.

Remarks

A geometric question is a question about a stopping rule. If the number of trials is fixed in advance, the binomial section is the right one; if the experiment stops at the first success, the geometric section is the right one. Memorylessness is the property that makes a fixed retry budget easy to analyse, and it is also the reason a failed attempt does not buy a higher chance on the next one.

8 Hypothesis test for one mean

A one-mean test asks whether a sample is compatible with a stated population mean μ_0. The claim is written as two hypotheses. H_0 always contains the boundary value, and H_1 is the departure the experiment was built to detect.

If the population standard deviation σ is known, or the sample is large enough that the sample standard deviation s can stand in for it, the statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}. \tag{12}$$

If σ is unknown and the sample is small, and the measurements are modelled as approximately normal, the statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1. \tag{13}$$

The test rejects H_0 when the statistic falls in the tail named by H_1. For a two-sided test at level α, the critical values are the $\alpha/2$ points. For a one-sided test they are the α points. Appendix C and the normal critical values in the last appendix are the reference.

A confidence interval carries the same arithmetic. The two-sided 95% interval for a known σ is

$$\bar{x} \pm 1.960\,\frac{\sigma}{\sqrt{n}}.$$

The interval contains μ_0 exactly when the two-sided 5% test does not reject $H_0 : \mu = \mu_0$.

Throughout this section the sample is treated as a random sample from the population named in the problem. A small t test also uses the modelling assumption that the population is approximately normal.

Problem 8.1. Claimed API latency

A vendor claims that the mean latency of an API is 50 milliseconds, with known standard deviation 12 milliseconds. A sample of 36 calls has mean 53 milliseconds. At the 5% level, test whether the mean is different from 50.

Solution. The hypotheses are $H_0 : \mu = 50$ and $H_1 : \mu \neq 50$, with $\alpha = 0.05$ and a two-sided z test. The standard error is $12/\sqrt{36} = 12/6 = 2$, so

$$z = \frac{53 - 50}{2} = 1.50.$$

The critical values are ± 1.960. Since $|1.50| < 1.960$, do not reject H_0. Appendix B gives $\Phi(1.50) = 0.9332$, so the two-sided p-value is

$$p = 2(1 - 0.9332) = 0.1336.$$

Answer. Do not reject H_0 ($z = 1.50$, $p = 0.1336$). The sample does not show, at the 5% level, that the mean latency differs from 50 ms.

Problem 8.2. Defect rate above a limit

A release rule says that the mean number of defects per package is 4, and a package line is investigated if there is evidence that the mean is higher. The population standard deviation is 2. A sample of 64 packages has mean 4.5. Test at the 5% level.

Solution. The investigation looks only for an increase, so $H_0 : \mu = 4$ and $H_1 : \mu > 4$. This is a one-sided z test at $\alpha = 0.05$. The standard error is $2/\sqrt{64} = 2/8 = 0.25$, and

$$z = \frac{4.5 - 4}{0.25} = 2.00.$$

The critical value is 1.645. Since $2.00 > 1.645$, reject H_0. The p-value is

$$p = 1 - \Phi(2.00) = 1 - 0.9772 = 0.0228.$$

Answer. Reject H_0 ($z = 2.00$, $p = 0.0228$). At the 5% level there is evidence that the mean number of defects per package is above 4.

Problem 8.3. A faster page load

The historical mean page-load time is 100 milliseconds, with known standard deviation 15 milliseconds. After a change, 25 page loads have mean 94 milliseconds. Test, at the 5% level, whether the mean has decreased.

Solution. The alternative has a direction: $H_0 : \mu = 100$ and $H_1 : \mu < 100$. The standard error is $15/\sqrt{25} = 3$, and

$$z = \frac{94 - 100}{3} = -2.00.$$

The critical value is -1.645. Since $-2.00 < -1.645$, reject H_0. By symmetry the p-value equals the upper tail at 2.00:

$$p = 0.0228.$$

Answer. Reject H_0 ($z = -2.00$, $p = 0.0228$). The sample supports a decrease in mean page-load time at the 5% level.

Problem 8.4. Context-switch time, unknown variance

A kernel claim is that the mean context-switch time is 20 microseconds. The population standard deviation is unknown. Sixteen measured switches have mean 22 and sample standard deviation 4. The measurements are modelled as approximately normal. Test the claim against a two-sided alternative at the 5% level.

Solution. Here $H_0 : \mu = 20$, $H_1 : \mu \neq 20$, and the statistic is t on $\nu = 15$ degrees of freedom. The standard error is $4/\sqrt{16} = 1$, so

$$t = \frac{22 - 20}{1} = 2.000.$$

Appendix C gives $t_{0.025,15} = 2.131$. Since $|2.000| < 2.131$, do not reject H_0.

The same row of the table places the statistic between the one-sided 5% point 1.753 and the one-sided 2.5% point 2.131. The two-sided p-value therefore satisfies $0.05 < p < 0.10$.

Answer. Do not reject H_0 ($t = 2.000$ on 15 degrees of freedom). The data do not show a difference from 20 microseconds at the 5% level.

Problem 8.5. A small sample of scores

Ten students complete a quiz. The sample mean is 78 and the sample standard deviation is 6. The scores are modelled as approximately normal. Test $H_0 : \mu = 75$ against $H_1 : \mu > 75$ at the 5% level.

Solution. The degrees of freedom are 9. The standard error uses $\sqrt{10} = 3.1623$:

$$\frac{s}{\sqrt{n}} = \frac{6}{3.1623} = 1.8974, \qquad t = \frac{78 - 75}{1.8974} = 1.581.$$

Appendix C gives $t_{0.05,9} = 1.833$. Since $1.581 < 1.833$, do not reject H_0. The statistic lies between $t_{0.10,9} = 1.383$ and $t_{0.05,9} = 1.833$, so $0.05 < p < 0.10$.

Answer. Do not reject H_0 ($t = 1.581$ on 9 degrees of freedom). A mean of 78 on ten scripts is not, at the 5% level, evidence that the population mean exceeds 75.

Problem 8.6. A one-percent test

Twenty-five measurements of a queue length have mean 5.2 and sample standard deviation 1. The population is modelled as approximately normal. Test $H_0 : \mu = 5$ against $H_1 : \mu \neq 5$ at the 1% level.

Solution. The standard error is $1/\sqrt{25} = 0.2$, and

$$t = \frac{5.2 - 5}{0.2} = 1.000$$

on 24 degrees of freedom. A two-sided 1% test uses $t_{0.005,24} = 2.797$. Since $|1.000| < 2.797$, do not reject H_0.

The statistic is also smaller than the one-sided 10% point $t_{0.10,24} = 1.318$, so each tail probability exceeds 0.10 and the two-sided p-value exceeds 0.20.

Answer. Do not reject H_0 ($t = 1.000$ on 24 degrees of freedom). The mean 5.2 is compatible with a population mean of 5 at the 1% level.

Problem 8.7. A large job sample

A scheduler is supposed to have mean job time 500 milliseconds. One hundred jobs have mean 510 and sample standard deviation 40. Because the sample is large, use s in place of σ and a z test. Test $H_0 : \mu = 500$ against $H_1 : \mu > 500$ at the 5% level.

Solution. The standard error is $40/\sqrt{100} = 4$, so

$$z = \frac{510 - 500}{4} = 2.50.$$

The one-sided 5% critical value is 1.645. Since $2.50 > 1.645$, reject H_0. Appendix B gives

$$p = 1 - \Phi(2.50) = 1 - 0.9938 = 0.0062.$$

The excess is 10 milliseconds, a quarter of the sample standard deviation of a single job, but the standard error of the mean is only 4 milliseconds. A modest shift can be detected when the sample is large.

Answer. Reject H_0 ($z = 2.50$, $p = 0.0062$). At the 5% level the mean job time is higher than 500 ms.

Problem 8.8. A strong increase on a small sample

Nine runs of a compression tool have mean time 18 seconds and sample standard deviation 3 seconds. The runs are modelled as approximately normal. The historical mean is 15 seconds. Test $H_0 : \mu = 15$ against $H_1 : \mu > 15$, first at the 5% level and then at the 1% level.

Solution. The standard error is $3/\sqrt{9} = 1$, so $t = (18 - 15)/1 = 3.000$ on 8 degrees of freedom.

- At 5%, the critical value is $t_{0.05,8} = 1.860$. Since $3.000 > 1.860$, reject H_0.
- At 1%, the critical value is $t_{0.01,8} = 2.896$. Since $3.000 > 2.896$, reject H_0 at this stricter level as well.

The next column is $t_{0.005,8} = 3.355$. Because $2.896 < 3.000 < 3.355$, the one-sided p-value satisfies $0.005 < p < 0.01$.

Answer. Reject H_0 at both 5% and 1% ($t = 3.000$ on 8 degrees of freedom, with $0.005 < p < 0.01$).

Problem 8.9. Confidence interval and two tests

Thirty-six response times have mean 120 milliseconds. The population standard deviation is known to be 18 milliseconds.

1. Construct a 95% confidence interval for the mean.
2. Test $H_0 : \mu = 115$ against $H_1 : \mu \neq 115$ at the 5% level.
3. Test $H_0 : \mu = 110$ against $H_1 : \mu \neq 110$ at the 5% level.

Explain the connection between the interval and the two decisions.

Solution. The standard error is $18/\sqrt{36} = 3$.

1. The 95% margin is $1.960 \times 3 = 5.880$, so the interval is
$$120 \pm 5.880 = (114.120,\ 125.880).$$
2. $$z = \frac{120 - 115}{3} = 1.667,$$
which rounds to 1.67 for the table. The critical values are ± 1.960. Since $|1.67| < 1.960$, do not reject H_0. Also $\Phi(1.67) = 0.9525$, so
$$p = 2(1 - 0.9525) = 0.0950.$$
3. $$z = \frac{120 - 110}{3} = 3.333,$$
which rounds to 3.33. Since $3.33 > 1.960$, reject H_0. Appendix B gives $\Phi(3.33) = 0.9996$, so
$$p = 2(1 - 0.9996) = 0.0008.$$

The value 115 lies inside (114.120, 125.880), and the two-sided 5% test does not reject it. The value 110 lies outside the interval, and the same test rejects it. For a known σ, the interval and the two-sided test are the same arithmetic.

Answer. (a) (114.12, 125.88). (b) Do not reject $\mu = 115$ ($z = 1.67$, $p = 0.0950$). (c) Reject $\mu = 110$ ($z = 3.33$, $p = 0.0008$).

Problem 8.10. Rejection region on the original scale

A team will claim that mean latency has increased if a one-sided 5% test rejects $H_0 : \mu = 50$ in favour of $H_1 : \mu > 50$. The population standard deviation is 10 milliseconds and the planned sample size is 16.

1. Find the sample means that lead to rejection.
2. A sample then returns $\bar{x} = 56$. Carry out the test and give the p-value.
3. Say what a Type I error is in this experiment, and give its probability when the mean is really 50.

Solution. The standard error is $10/\sqrt{16} = 2.5$.

1. Reject H_0 when $z > 1.645$, that is, when
$$\bar{x} > 50 + 1.645 \times 2.5 = 50 + 4.1125 = 54.1125.$$
Rounded to two decimal places, the rejection region is $\bar{x} > 54.11$.
2. The observed mean 56 is inside that region.
$$z = \frac{56 - 50}{2.5} = 2.40, \qquad p = 1 - \Phi(2.40) = 1 - 0.9918 = 0.0082.$$
Reject H_0.
3. A Type I error is rejecting H_0 when the mean latency is still 50 milliseconds: the team announces an increase that is not there. The test was built so that this probability is $\alpha = 0.05$.

Answer. (a) Reject when the sample mean exceeds 54.11 ms. (b) Reject H_0 ($z = 2.40$, $p = 0.0082$). (c) A Type I error is declaring an increase when the true mean is still 50 ms; its probability is 0.05.

Remarks

The size of $\bar{x} - \mu_0$ is not the test. Problem 8.7 rejects on a shift of 10 milliseconds because the standard error is 4; Problem 8.5 does not reject on a shift of 3 marks because the standard error is nearly 1.9 and the sample has only ten observations. The critical value also depends on the direction that was promised before seeing the data. A one-sided 5% test uses 1.645, not 1.960, and it does not entitle the experimenter to switch direction after looking at the sign of $\bar{x} - \mu_0$.

9 Hypothesis test for two means

A two-mean test compares two populations rather than a population and a constant. Write $\delta = \mu_1 - \mu_2$. The null hypothesis in this section is $\delta = 0$, the statement that the two means are equal. The samples are independent except in the paired problem, where the same benchmark is measured twice and the analysis is performed on the differences.

When both population standard deviations are known,

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}. \tag{14}$$

When they are unknown but treated as equal, the pooled variance is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \tag{15}$$

and

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}, \qquad \nu = n_1 + n_2 - 2. \tag{16}$$

For a large sample, s_1 and s_2 may be inserted directly into the z formula. For paired data $d_i = x_{i1} - x_{i2}$, the test is the one-mean t test of Section 8 applied to the differences, with $H_0 : \mu_d = 0$.

The 95% interval for a difference with known standard errors of the kind in the z formula is

$$(\bar{x}_1 - \bar{x}_2) \pm 1.960\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}.$$

Zero lies in that interval exactly when the two-sided 5% test does not reject equal means.

Problem 9.1. Two schedulers

Scheduler A is measured on 50 workloads, with mean completion time 30 and known standard deviation 5. Scheduler B is measured on 50 workloads, with mean 28 and known standard deviation 5. The samples are independent. Test, at the 5% level, whether the two mean completion times differ.

Solution. The hypotheses are $H_0 : \mu_A = \mu_B$ and $H_1 : \mu_A \neq \mu_B$. The standard error of the difference is

$$\sqrt{\frac{25}{50} + \frac{25}{50}} = \sqrt{1} = 1,$$

so

$$z = \frac{30 - 28}{1} = 2.00.$$

The critical values are ± 1.960. Since $2.00 > 1.960$, reject H_0. The two-sided p-value is

$$p = 2(1 - \Phi(2.00)) = 2 \times 0.0228 = 0.0456.$$

The result is significant at 5%, but only just: 0.0456 is close to the threshold.

Answer. Reject equal means ($z = 2.00$, $p = 0.0456$). At the 5% level the mean completion times differ.

Problem 9.2. Two CPUs

Under the same benchmark, CPU A has sample size 40, mean temperature 62, and known standard deviation 8. CPU B has sample size 40, mean 58, and known standard deviation 8. The samples are independent. Test, at the 5% level, whether the mean temperature of CPU A is higher.

Solution. The alternative is one-sided: $H_0 : \mu_A = \mu_B$ and $H_1 : \mu_A > \mu_B$.

$$\sqrt{\frac{64}{40} + \frac{64}{40}} = \sqrt{3.2} = 1.7889,$$

$$z = \frac{62 - 58}{1.7889} = 2.236.$$

The critical value is 1.645. Since $2.236 > 1.645$, reject H_0. For the table, 2.236 rounds to 2.24, and

$$p \approx 1 - \Phi(2.24) = 1 - 0.9875 = 0.0125.$$

Answer. Reject H_0 ($z = 2.236$, table $p \approx 0.0125$). At the 5% level CPU A runs hotter than CPU B.

Problem 9.3. Equal-sized benchmark samples

Two implementations are timed on independent samples of 16 programs each. Implementation A has mean 42 and sample standard deviation 4. Implementation B has mean 38 and sample standard deviation 4. Assume equal population variances and approximate normality. Test whether the means differ at the 5% level.

Solution. The sample standard deviations agree, so the pooled value is $s_p = 4$. The degrees of freedom are $16 + 16 - 2 = 30$. The standard error of the difference is

$$s_p\sqrt{\frac{1}{16} + \frac{1}{16}} = 4\sqrt{\frac{1}{8}} = \frac{4}{\sqrt{8}} = \sqrt{2} = 1.4142.$$

The test statistic is

$$t = \frac{42 - 38}{1.4142} = 2.828,$$

which is also $\sqrt{8}$. A two-sided 5% test uses $t_{0.025,30} = 2.042$. Since $2.828 > 2.042$, reject H_0. The statistic also exceeds $t_{0.005,30} = 2.750$, so the two-sided p-value is below 0.01.

Answer. Reject equal means ($t = 2.828$ on 30 degrees of freedom, two-sided $p < 0.01$).

Problem 9.4. Two compilers, unequal samples

Compiler A is assessed on 10 programs: the mean code size is 85 and the sample standard deviation is 6. Compiler B is assessed on 12 programs: the mean is 78 and the sample standard deviation is 5. Assume equal population variances and approximate normality. Test whether the mean code sizes differ at the 5% level.

Solution. The pooled variance uses $\nu = 10 + 12 - 2 = 20$:

$$s_p^2 = \frac{9 \times 6^2 + 11 \times 5^2}{20} = \frac{9 \times 36 + 11 \times 25}{20} = \frac{324 + 275}{20} = \frac{599}{20} = 29.95.$$

$$s_p = \sqrt{29.95} = 5.4727.$$

$$\sqrt{\frac{1}{10} + \frac{1}{12}} = \sqrt{\frac{11}{60}} = 0.4282,$$

but the standard error below keeps the square root unrounded inside the product:

$$SE = 5.4727\sqrt{\frac{11}{60}} = 2.3433, \qquad t = \frac{85 - 78}{2.3433} = 2.987.$$

The two-sided 5% critical value is $t_{0.025,20} = 2.086$. Since $2.987 > 2.086$, reject H_0. Also $t_{0.005,20} = 2.845$, and $2.987 > 2.845$, so the two-sided p-value is below 0.01.

Answer. Reject equal means ($t = 2.987$ on 20 degrees of freedom, two-sided $p < 0.01$).

Problem 9.5. Large samples of response times

Service A is observed for 100 requests: the mean response is 250 milliseconds and the sample standard deviation is 20. Service B is observed for 100 requests: the mean is 242 and the sample standard deviation is 18. The samples are independent. Using a large-sample z test, test whether the means differ at the 1% level.

Solution.

$$\sqrt{\frac{20^2}{100} + \frac{18^2}{100}} = \sqrt{4 + 3.24} = \sqrt{7.24} = 2.6907.$$

$$z = \frac{250 - 242}{2.6907} = 2.973.$$

A two-sided 1% test uses ± 2.576. Since $2.973 > 2.576$, reject H_0. For the table, 2.973 rounds to 2.97, and $\Phi(2.97) = 0.9985$, so

$$p = 2(1 - 0.9985) = 0.0030.$$

Answer. Reject equal means at the 1% level ($z = 2.973$, table $p = 0.0030$).

Problem 9.6. A small gap that is not significant

Two caches are compared on independent samples of 25 lookups each. Cache A has mean access time 10, and cache B has mean 11. Both population standard deviations are known and equal to 3. Test whether the means differ at the 5% level.

Solution.

$$\sqrt{\frac{9}{25} + \frac{9}{25}} = \sqrt{0.72} = 0.8485, \qquad z = \frac{10 - 11}{0.8485} = -1.179.$$

The critical values are ± 1.960. Since $|-1.179| < 1.960$, do not reject H_0. For the table, 1.179 rounds to 1.18, and $\Phi(1.18) = 0.8810$, so

$$p = 2(1 - 0.8810) = 0.2380.$$

A one-unit gap is real as a description of these two samples, and it is still compatible with equal population means. The standard error of the difference is 0.85, so a gap of 1 is a little more than one standard error.

Answer. Do not reject equal means ($z = -1.179$, table $p = 0.2380$).

Problem 9.7. Paired benchmark times

An optimisation is timed on the same eight benchmarks before and after the change. The times, in seconds, are:

Benchmark	1	2	3	4	5	6	7	8
Before	20	18	24	16	30	15	22	19
After	15	15	20	14	24	14	18	16

Let d be the before-minus-after difference. Test $H_0 : \mu_d = 0$ against $H_1 : \mu_d > 0$ at the 5% level. A positive mean difference means the optimisation reduced the runtime. Model the differences as approximately normal.

Solution. The eight differences are $5, 3, 4, 2, 6, 1, 4, 3$. Their sum is 28, so $\bar{d} = 28/8 = 3.5$. The deviations from 3.5 are

$$1.5,\ -0.5,\ 0.5,\ -1.5,\ 2.5,\ -2.5,\ 0.5,\ -0.5,$$

and the sum of their squares is

$$2.25 + 0.25 + 0.25 + 2.25 + 6.25 + 6.25 + 0.25 + 0.25 = 18.$$

The sample variance of the differences is $s_d^2 = 18/7 = 2.5714$. The t statistic can be computed without a further rounding step:

$$t = \frac{3.5}{\sqrt{18/56}} = \frac{3.5}{\sqrt{9/28}} = \frac{3.5}{3/\sqrt{28}} = \frac{7}{6}\sqrt{28}.$$

Since $\sqrt{28} = 5.2915$,

$$t = \frac{7}{6} \times 5.2915 = 6.173$$

on 7 degrees of freedom. The one-sided 5% critical value is $t_{0.05,7} = 1.895$. Since $6.173 > 1.895$, reject H_0. The statistic also exceeds $t_{0.005,7} = 3.499$, so $p < 0.005$.

Pairing is what makes this result sharp. The benchmarks differ from one another by more than the optimisation does, but those program-to-program differences cancel in d.

Answer. Reject H_0 ($t = 6.173$ on 7 degrees of freedom, $p < 0.005$). The mean reduction of 3.5 seconds is significant at the 5% level.

Problem 9.8. Confidence interval for the same gap

Return to the two schedulers of Problem 9.1: the difference of sample means is 2, and the standard error of that difference is 1.

1. Construct a 95% confidence interval for $\mu_A - \mu_B$.
2. Construct a 99% confidence interval for the same difference.
3. Say which of the two intervals contains 0, and what that implies for two-sided tests at 5% and at 1%.

Solution.

1. The 95% interval is
$$2 \pm 1.960 \times 1 = (0.040,\ 3.960).$$
2. The 99% interval uses the two-sided 1% point 2.576:
$$2 \pm 2.576 = (-0.576,\ 4.576).$$
3. Zero is outside the 95% interval and inside the 99% interval. Therefore a two-sided test rejects equal means at the 5% level, which is the decision already made in Problem 9.1, and it does not reject equal means at the 1% level. The p-value 0.0456 sits between 0.01 and 0.05, so the two intervals were bound to disagree.

Answer. (a) (0.040, 3.960). (b) (−0.576, 4.576). (c) Only the 95% interval excludes 0: significant at 5%, not at 1%.

Problem 9.9. A new queue that is not yet shown to be faster

A new queue is compared with the old queue on independent samples of 12 waiting times each. The new queue has mean 30 and sample standard deviation 5. The old queue has mean 33 and sample standard deviation 5. Assume equal variances and approximate normality. Test, at the 5% level, whether the new queue has the smaller mean.

Solution. The alternative is $H_1 : \mu_{\text{new}} < \mu_{\text{old}}$. The pooled standard deviation is $s_p = 5$, and the degrees of freedom are 22.

$$SE = 5\sqrt{\frac{1}{12} + \frac{1}{12}} = 5\sqrt{\frac{1}{6}}.$$

Using $\sqrt{1/6} = 0.4082$,

$$SE = 5 \times 0.4082 = 2.0410, \qquad t = \frac{30 - 33}{2.0410} = -1.470.$$

The critical value is $-t_{0.05,22} = -1.717$. Since -1.470 is not less than -1.717, do not reject H_0. The absolute statistic lies between $t_{0.10,22} = 1.321$ and $t_{0.05,22} = 1.717$, so $0.05 < p < 0.10$.

The sample means do point the right way: 30 is less than 33. The test says that a gap of 3, with this standard error and these sample sizes, is not yet significant at 5%.

Answer. Do not reject equal means ($t = -1.470$ on 22 degrees of freedom, $0.05 < p < 0.10$).

Problem 9.10. A result that depends on the significance level

Two caches are compared on independent samples of 36 requests each. Cache A has mean hit time 12.4 and cache B has mean 11.2. Both population standard deviations are known and equal to 2. Test whether the means differ, first at the 5% level and then at the 1% level, and give the two-sided p-value from the normal table.

Solution.

$$\sqrt{\frac{4}{36} + \frac{4}{36}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3} = \frac{1.4142}{3} = 0.4714.$$

$$z = \frac{12.4 - 11.2}{0.4714} = 2.546.$$

- At 5%, the critical value is 1.960. Since $2.546 > 1.960$, reject equal means.
- At 1%, the critical value is 2.576. Since $2.546 < 2.576$, do not reject equal means.

For the table, 2.546 rounds to 2.55, and $\Phi(2.55) = 0.9946$, so

$$p = 2(1 - 0.9946) = 0.0108.$$

The p-value lies between 0.01 and 0.05. Changing the significance level from 5% to 1% changes the decision, and the interval arithmetic of Problem 9.8 is the same phenomenon.

Answer. Reject equal means at 5%, and do not reject them at 1%. The table p-value is 0.0108, from $z = 2.546$.

Remarks

An independent two-sample test spends its sample on two separate sources of noise. A paired test spends it on the difference, which is why the eight benchmarks in Problem 9.7 produce a t statistic of 6.173 from a mean reduction of 3.5 seconds. Pairing is available only when the two numbers on a row are measurements of the same unit. Pooling the two compilers is a variance assumption, not a consequence of the t table: it was reasonable there because the sample standard deviations, 6 and 5, were close, and the problem said to assume equality.

The last two problems are the practical warning. A difference can be large enough for a 5% test and too small for a 1% test. The p-value reports that position; the significance level only converts it into a yes or a no.

10 Quick answer key

The worked solution above each problem is the record of the reasoning. This list is only the final reported value.

Problem	Answer
1.1	0.1937
1.2	0.3671
1.3	mean 6, variance 4.2, standard deviation 2.0494
1.4	0.7373
1.5	0.2621 all succeed; 0.7379 at least one failure
1.6	0.2581
1.7	0.3456
1.8	expected clicks 3; $P(X = 0) = 0.0352$

Problem	Answer
1.9	0.9012
1.10	0.5613
2.1	$P(X = 2) = 0.2240$; $P(X = 0) = 0.0498$
2.2	0.2231 none; 0.7769 at least one
2.3	0.6703 none; 0.3297 at least one
2.4	$P(X = 4) = 0.1954$; $P(X < 3) = 0.2381$
2.5	(a) 0.3233; (b) 0.1339
2.6	0.8088
2.7	$P(X = 4) = P(X = 5) = 0.1755$; standard deviation 2.2361
2.8	(a) 0.9179; (b) mean 10, $P(Y = 0) = 0.000045$
2.9	(a) 0.8187 and 0.1637; (b) 0.1804
2.10	mean 6, variance 6, standard deviation 2.4495; $P(X \leq 2) = 0.0620$
3.1	0.8413
3.2	0.0228
3.3	0.9544
3.4	83.16
3.5	0.0228
3.6	0.8185
3.7	0.0668
3.8	0.9544
3.9	96.45 ms
3.10	both probabilities 0.1056
4.1	0.1804
4.2	0.1353
4.3	0.4060
4.4	0.1954
4.5	0.9502
4.6	0.2240
4.7	0.3233
4.8	0.1008
4.9	exact 0.3697; Poisson 0.3679; absolute error 0.0018
4.10	mean 2, variance 2; $P(X \geq 4) \approx 0.1429$
5.1	0.7286
5.2	0.0735
5.3	0.0256
5.4	0.0796
5.5	0.7062
5.6	0.0202
5.7	0.0179
5.8	normal approximation invalid; Poisson 0.2707
5.9	0.6482
5.10	0.0016
6.1	0.0527
6.2	0.1875
6.3	mean 20, variance 80, standard deviation 8.9443
6.4	mean 20; $P(X = 5) = 0.0292$
6.5	0.3277; mean 6.25
6.6	0.1792
6.7	0.1080
6.8	0.1875; mean failures 3; same event as $X = 5$
6.9	0.1792
6.10	one success: mean 4, variance 12; four successes: mean 16, variance 48
7.1	0.1024; mean 5, variance 20, standard deviation 4.4721
7.2	0.875
7.3	0.5905

Problem	Answer
7.4	0.5882
7.5	0.5625
7.6	mean 20, standard deviation 19.4936; $P(X = 1) = 0.05$
7.7	A: mean 2.5, sd 1.9365; B: mean 1.25, sd 0.5590
7.8	0.0960; 0.1600; mean 1.6667
7.9	$k = 9$
7.10	mean 16 ms; $P(T > 16) = 0.3164$
8.1	do not reject; $z = 1.50$, $p = 0.1336$
8.2	reject; $z = 2.00$, $p = 0.0228$
8.3	reject; $z = -2.00$, $p = 0.0228$
8.4	do not reject; $t = 2.000$, $0.05 < p < 0.10$
8.5	do not reject; $t = 1.581$, $0.05 < p < 0.10$
8.6	do not reject; $t = 1.000$, two-sided $p > 0.20$
8.7	reject; $z = 2.50$, $p = 0.0062$
8.8	reject at 5% and at 1%; $t = 3.000$, $0.005 < p < 0.01$
8.9	interval $(114.12, 125.88)$; do not reject 115; reject 110
8.10	reject for $\bar{x} > 54.11$; observed $z = 2.40$, $p = 0.0082$; Type I probability 0.05
9.1	reject; $z = 2.00$, $p = 0.0456$
9.2	reject; $z = 2.236$, table $p \approx 0.0125$
9.3	reject; $t = 2.828$, two-sided $p < 0.01$
9.4	reject; $t = 2.987$, two-sided $p < 0.01$
9.5	reject; $z = 2.973$, table $p = 0.0030$
9.6	do not reject; $z = -1.179$, table $p = 0.2380$
9.7	reject; $t = 6.173$, $p < 0.005$
9.8	95% interval $(0.040, 3.960)$; 99% interval $(-0.576, 4.576)$
9.9	do not reject; $t = -1.470$, $0.05 < p < 0.10$
9.10	reject at 5%, not at 1%; table $p = 0.0108$

A Values of $e^{-\lambda}$

The Poisson solutions use these values, rounded half up to six decimal places. A final probability is then rounded half up to four decimal places.

λ	$e^{-\lambda}$	λ	$e^{-\lambda}$
0.2	0.818731	0.4	0.670320
0.5	0.606531	0.8	0.449329
1	0.367879	1.5	0.223130
2	0.135335	2.4	0.090718
2.5	0.082085	3	0.049787
4	0.018316	5	0.006738
6	0.002479	10	0.000045

B Standard normal table

The row label is the units and tenths of z. The column label is the hundredths. The entry is $\Phi(z) = P(Z \le z)$, rounded half up to four decimal places. For a negative value use $\Phi(-z) = 1 - \Phi(z)$. For example, the row 1.2 and the column .05 give $\Phi(1.25) = 0.8944$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

C Critical values of Student's t

Each entry is the upper-tail critical value $t_{\alpha,\nu}$: the number c for which $P(T > c) = \alpha$ when T has ν degrees of freedom. Round a computed t to three decimal places before comparing. The last row is the standard normal critical value, the limit as $\nu \to \infty$. A two-sided test at level α uses the one-sided column $\alpha/2$. In particular, a two-sided 5% test uses the column 0.025, and a two-sided 1% test uses the column 0.005.

ν	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819

ν	0.10	0.05	0.025	0.01	0.005
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

D Normal critical values used in the tests

These are the usual critical points of Z, consistent with the last row of the t table.

Tail probability	Critical value	Typical use
0.05	1.645	one-sided test at 5%
0.025	1.960	two-sided test at 5%; 95% confidence interval
0.01	2.326	one-sided test at 1%
0.005	2.576	two-sided test at 1%; 99% confidence interval